

ADDENDUM · CRITICAL PEER REVIEW

Variance Retraction, Tilted-Pauli Split, and Status Corrections

Resolution of the strength-measure question, restoration of the $\omega(J^2)$ construction, and honest documentation of retractions and open targets

SUPPLEMENT TO MAJOR REVISION

Independent Reviewer

Supplement to the RCF v2.1 / v3.0 Critical Review

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1 Executive Summary

1.1 Purpose

This document supplements the critical peer review of the Reconciliation Causal Framework (RCF) v2.1 / v3.0. It records the resolution of a methodological question that arose during follow-up analysis: whether the framework’s mismatch strength measure $\omega(J_{\alpha\beta}^2)$ should be replaced by the variance form $\text{Var}_\omega(J_{\alpha\beta}) = \omega(J_{\alpha\beta}^2) - \omega(J_{\alpha\beta})^2$.

The variance substitution was explored as a candidate replacement. It produced results that contradict the framework’s central homogeneity claim. The correct response is to retract the candidate, not to rewrite the result. This addendum documents that retraction, restores the original $\omega(J^2)$ construction, and updates the framework’s honest status accordingly.

The original $\omega(J^2)$ construction (v3.0 Definition 1.17) is restored as the framework’s primary mismatch strength measure. The variance substitution is retracted. The Pauli-zero ($\hat{F}_{MM} = 0$ for pure Pauli constraints) is reinterpreted as the “fully reconciled limit” — a physical prediction that a universe with Pauli constraints would have no dynamics, rather than a defect in the construction.

1.2 What is preserved

- The $\hat{F}_{MM} = \hat{F}_{MM}^{(0)} + \hat{F}_{MM}^{(1)}[\rho]$ decomposition for tilted-Pauli constraints, with critic-verified arithmetic (Section 4).
- The Pauli-zero as a legitimate physical prediction: pure Pauli constraints give $\hat{F}_{MM} = 0$ identically — no RP flow, no dynamics, no dark sector (Section 3). Verified robust under multi-sector extension (Section 5).
- The dark-energy fingerprint as a *structural* prediction: observed $\rho_{DE} > 0$ implies the universe’s constraints are non-Pauli ($\theta \neq 0$). The empirical form remains conditional on T-2/T-3 (Section 6).
- The hierarchical homogeneity proof on its original footing: the linear driver $\hat{F}_{MM}^{(1)} = \vec{W} \cdot \vec{r}$ is present under $\omega(J^2)$ and provides the constraint plane the hierarchical proof requires (Section 7).
- **Single-sector vs multi-sector scoping** (new): the tilted-Pauli calculation is a local intra-sector result; cosmological interpretations require the multi-sector structure (Section 5).

1.3 What is retracted

- The variance substitution $\omega(J^2) \rightarrow \text{Var}_\omega(J)$: not derived from framework primitives; contradicts the homogeneity claim.
- The “Mexican-hat” / spontaneous-symmetry-breaking framing: a contradiction dressed as a feature, not a derived result.
- The $n_s \approx 0.667$ prediction: arithmetically wrong ($n_s = 2/3$, off by 0.3 from Planck’s $n_s \approx 0.965$).

- The “three seeds of structural mismatch” framing (Jacobi is an identity; Kaplansky is eligibility; the third was unverifiable).
- The $E = mc^2$ derivation from “resistance to reconciliation”: blocked on Lorentz invariance and four-momentum construction.
- All numerical $|\vec{r}|_*$ values: three different numbers from the same formula, none verified.
- **Global cosmological overextensions of the single-sector calculation** (new): interpreting the intra-sector tilted-Pauli result as a global dark-energy source, global homogeneity proof, or fluctuation spectrum. These require the multi-sector structure (Section 5).

1.4 What remains open

- The layer-hierarchical RP principle (bridge from definitional acyclicity to dynamical hierarchy is unbuilt).
- T-2 (spectral gap + bounded degree), T-3 (continuum limit), and the Friedmann derivation (GAP-6.2.2).
- The dimensional connection from algebraic $\hat{F}_{MM}^{(0)}$ to cosmological ρ_{DE} .
- The fluctuation spectrum ($\delta T/T$, n_s , r , Gaussianity).

1.5 Primitive count

The primitive count is **5** (relational algebra, constraints, Lie–Jordan decomposition, Master-Zero, inflation-replacement requirement), *conditional on 4* if the hierarchical principle is proven. The unconditional claim of primitive count 4 is retracted.

2 Resolution of the Variance Question

2.1 The candidate substitution

During follow-up analysis, a candidate replacement for the correlational strength measure was considered:

$$\omega(J_{\alpha\beta}^2) \longrightarrow \text{Var}_\omega(J_{\alpha\beta}) = \omega(J_{\alpha\beta}^2) - \omega(J_{\alpha\beta})^2.$$

The motivation was the intuition that “a state-independent correlation should count as reconciled, not as no correlation.” For pure Pauli constraints, $J_{ab} = \delta_{ab}\mathbf{1}$ is central, so $\omega(J_{ab}^2) = \delta_{ab}$ and $\hat{F}_{MM} = 0$ identically. The variance substitution was motivated by making this vanishing “correct by construction.”

2.2 Why the substitution is rejected

1. **Not derived from framework primitives.** The strength measure $\omega(J_{\alpha\beta}^2)$ is the squared norm of the Jordan product in the GNS Hilbert space — the natural bilinear, non-negative, symmetric pairing derived from $AB = \frac{1}{2}[A, B] + A \circ B$. Replacing it with variance requires an independent justification, and none was established.
2. **Contradicts the central homogeneity claim.** When applied to tilted-Pauli constraints, the variance substitution eliminates the linear driver $\hat{F}_{MM}^{(1)} = \vec{W} \cdot \vec{r}$ entirely (verified by symmetry: the sextet of ordered pairs with $\alpha \neq \beta$ is fully permutation-symmetric, so no term linear in a single r_i survives). It also introduces a negative quadratic $\hat{F}_{MM}^{(2)} \leq 0$ that drives

the RP flow *away* from $\vec{r} = 0$. The minimizer under variance is at $|\vec{r}|_* > 0$, contradicting the homogeneity derivation. The correct response to a candidate substitution that contradicts a core result is to discard the substitution, not rewrite the result.

3. **The Pauli-zero is already correct without variance.** Under $\omega(J^2)$, the Pauli-zero is a legitimate physical prediction (Section 3). The variance path adds nothing.

2.3 The methodological lesson

The variance episode is a case study in the pattern the critic has identified repeatedly: a mathematically correct intermediate result was over-interpreted, layered with un-derived physical framing (“spontaneous symmetry breaking,” “Mexican-hat VEV”), and presented as progress when it contradicted a core result. The n_s prediction derived from the variance path was arithmetically wrong ($n_s = 2/3$, off by 0.3 from Planck’s 0.965) and was presented as “testable against” the data when the numbers did not agree.

Status: Option (a) adopted. The original $\omega(J^2)$ construction is restored as primary. The variance path is retracted. All claims resting on the variance substitution are withdrawn (Section 8).

3 Restored $\omega(J^2)$ Construction

3.1 Primary form

The framework’s mismatch strength measure is restored to its original v3.0 form (Definition 1.17):

Definition 3.1 (Lie–Jordan Mismatch Operator — Primary Form). Let $\{\hat{C}_\alpha\}_{\alpha \in I}$ be the constraint set, $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ the Lie obstruction, and $J_{\alpha\beta} = \hat{C}_\alpha \circ \hat{C}_\beta$ the Jordan co-occurrence. For a state ω on $\mathcal{A}$, define:

$$\hat{F}_{MM} := \sum_{\alpha, \beta \in I} \omega(J_{\alpha\beta}^2) \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta}).$$

The corresponding burden functional is $B_{MM}[\rho] := \text{Tr}(\rho \hat{F}_{MM})$.

3.2 Properties (unchanged from v3.0)

- **Positivity:** $\hat{F}_{MM} \geq 0$, since each summand is a positive operator weighted by a non-negative scalar.
- **Reduction to $\hat{F}$:** in the uniform-Jordan limit ($\omega(J_{\alpha\beta}^2) = c$ for all Jordan-adjacent pairs), $\hat{F}_{MM} = c \cdot \hat{F}_{\alpha \sim \beta}$ (the Jordan-restricted fracture operator).
- **State-zero on no Jordan structure:** if $\omega(J_{\alpha\beta}^2) = 0$ for all α, β , then $\hat{F}_{MM} = 0$.

3.3 The Pauli-zero as the fully reconciled limit

For pure Pauli constraints $\hat{C}_a = \sigma_a$ ($a = 1, 2, 3$):

- Jordan products: $J_{ab} = \sigma_a \circ \sigma_b = \delta_{ab} \mathbf{1}$ (central, state-independent).
- $\omega(J_{ab}^2) = \delta_{ab}$ regardless of state.
- Lie obstructions: $\Delta_{ab} = 2i\epsilon_{abc}\sigma_c$, so $\rho(\Delta_{ab}^\dagger \Delta_{ab}) = 4(1 - \delta_{ab})$.

- For $a \neq b$: $\omega(J_{ab}^2) = 0$, so the summand vanishes.
- For $a = b$: $\Delta_{aa} = 0$, so the summand vanishes.
- **Result:** $\hat{F}_{MM} = 0$ identically, for all states ρ .

The Pauli-zero is not a defect in the construction. It is a physical prediction: a universe with pure Pauli constraints has no Lie–Jordan mismatch, no RP flow, no dynamics, and no dark sector. The Pauli constraints are the “fully reconciled limit” of the framework — the algebraic ground state where causality (Lie) and locality (Jordan) decouple completely. The observed universe has dynamics, so its constraints cannot be pure Pauli. This is the dark-energy fingerprint (Section 6).

3.4 What the Pauli-zero is *not*

- It is *not* a bug to be fixed by substituting variance.
- It is *not* an indication that the link construction is incomplete.
- It is *not* a sign that mismatch lives at higher order (triplets, paths, clusters).
- It *is* the correct answer for the right reason: pure Pauli constraints have no state-dependent Jordan structure, hence no mismatch to reconcile.

The earlier diagnosis that the link construction was “built incorrectly” because it gave zero for Pauli was a misreading. The construction is correct; the Pauli case is genuinely fully reconciled. The framework’s dynamics require non-Pauli constraints, which is a structural prediction, not a gap.

4 Critic-Corrected Tilted-Pauli Split

4.1 Setup

Tilted-Pauli constraints: $\hat{C}_\alpha = \cos \theta \sigma_\alpha + \sin \theta \mathbf{1}$, $\alpha \in \{1, 2, 3\}$. State: Bloch vector $\vec{r}$, $|\vec{r}| \leq 1$.

4.2 Structures ($\alpha \neq \beta$)

Using $\{\sigma_\alpha, \sigma_\beta\} = 0$ for $\alpha \neq \beta$:

$$\begin{aligned} J_{\alpha\beta} &= \hat{C}_\alpha \circ \hat{C}_\beta = \sin^2 \theta \mathbf{1} + \sin \theta \cos \theta (\sigma_\alpha + \sigma_\beta), \\ \Delta_{\alpha\beta} &= [\hat{C}_\alpha, \hat{C}_\beta] = 2i \cos^2 \theta \epsilon_{\alpha\beta\gamma} \sigma_\gamma, \quad \rho(\Delta_{\alpha\beta}^\dagger \Delta_{\alpha\beta}) = 4 \cos^4 \theta. \end{aligned}$$

4.3 Expectation and split

Writing $s = r_1 + r_2 + r_3$:

$$\omega(J_{\alpha\beta}^2) = \sin^4 \theta + 2 \sin^2 \theta \cos^2 \theta + 2 \sin^3 \theta \cos \theta (r_\alpha + r_\beta).$$

Summing over the 6 ordered pairs with $\alpha \neq \beta$:

$$\hat{F}_{MM}[\vec{r}] = 4 \cos^4 \theta \sum_{\alpha \neq \beta} \omega(J_{\alpha\beta}^2) = 24 \sin^2 \theta \cos^4 \theta (1 + \cos^2 \theta) + 32 \sin^3 \theta \cos^5 \theta \cdot s.$$

$$\hat{F}_{MM}[\vec{r}] = \hat{F}_{MM}^{(0)}(\theta) + \hat{F}_{MM}^{(1)}[\vec{r}]$$

$$\hat{F}_{MM}^{(0)} = 24 \sin^2 \theta \cos^4 \theta (1 + \cos^2 \theta)$$

$$\hat{F}_{MM}^{(1)} = 32 \sin^3 \theta \cos^5 \theta (r_1 + r_2 + r_3) = \vec{W} \cdot \vec{r}, \quad \vec{W} \propto (1, 1, 1)$$

$\hat{F}_{MM}^{(0)}$ is the state-independent irreducible floor (peak ≈ 6 at $\theta \approx 32.8^\circ$). $\hat{F}_{MM}^{(1)}$ is the linear driver.

Verification. Pauli ($\theta = 0$): $\hat{F}_{MM} = 0$ for all states $\checkmark$. Tilted ($\theta \neq 0$): $\hat{F}_{MM} > 0$, state-dependent $\checkmark$. At $\theta = \pi/4$: $\hat{F}_{MM}^{(0)} = 4.5$, $|\vec{W}| = 2\sqrt{3} \approx 3.46$, both $\mathcal{O}(1)$.

Note on earlier error. A previous draft gave $\hat{F}_{MM}^{(0)} = 24 \sin^4 \theta \cos^2 \theta$, which dropped the $|\vec{v}_{\alpha\beta}|^2$ contribution and miscomputed the commutator norm. The corrected formula above has been independently verified.

Scoping note. This calculation is a *local intra-sector* computation for a minimal $n = 2$ constraint set. It computes $\hat{F}_{MM}^{\text{intra}}$ for one sector. Cosmological interpretations (dark energy, global homogeneity, fluctuation spectrum) require the multi-sector structure, which this calculation does not provide. See Section 5 for the full scoping correction.

5 Single-Sector vs Multi-Sector Scoping

5.1 The scoping correction

The tilted-Pauli calculation (Section 4) is a **local intra-sector calculation**, not a global calculation. The cosmological interpretations attached to it (dark energy, global homogeneity, fluctuation spectrum) require the **multi-sector structure** that the framework's fracture (v3.0 §1.2) produces, which the single-sector calculation does not provide.

Single-sector (local): computes $\hat{F}_{MM}^{\text{intra}}$ for one minimal $n = 2$ constraint set. Valid as a statement about the internal structure of one sector.

Multi-sector (global): the full physical picture requires the fractured space $\mathcal{H}_{kin} \rightarrow \bigoplus_k \mathcal{H}_k \oplus \mathcal{H}_{cross}$, with total mismatch $\hat{F}_{MM}^{\text{total}} = \sum_k \hat{F}_{MM}^{(k)} + \hat{F}_{MM}^{\text{cross}}$. The single-sector calculation computes only the first sum.

5.2 Verification: Pauli-zero is robust under multi-sector extension

Structural reason. Pauli complementarity ($\{\sigma_a, \sigma_b\} = 2\delta_{ab}\mathbf{1}$; $[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c$) means: wherever the anticommutator is non-zero, the commutator is zero, and vice versa. This complementarity is preserved under tensor products and linear combinations, so it survives multi-sector extension.

Multi-sector test. For a two-sector system $M_4(\mathbb{C}) = M_2(\mathbb{C}) \otimes M_2(\mathbb{C})$ with sector-local Pauli constraints: intra-sector contributions are zero (same as single-sector); cross-sector commutators vanish (different tensor factors commute); cross-sector contribution is zero. **Total:** $\hat{F}_{MM} = 0$. Even with cross-coupling constraints, the Pauli complementarity ensures every pair has zero commutator or zero Jordan product.

Pauli-zero is **robust** under multi-sector extension. It holds identically for all states, in all configurations, with or without cross-sector coupling. This is a structural theorem of the Pauli algebra, not an artifact of the single-sector calculation.

5.3 What was overextended

- **Dark energy source:** Single-sector $F^{(0),k}$ sources particle mass, not dark energy. Dark energy lives in the cross-sector channel ($\hat{F}_{MM}^{(0),\text{cross}}$), not computed.
- **Global homogeneity:** The hierarchical proof (Section 7) is valid as a local statement. Cosmological homogeneity is a multi-sector statement requiring the sector field $\{\vec{r}_k\}$.
- **Fluctuation spectrum:** Fluctuations are deviations of the sector field from uniformity. A single sector has nothing to fluctuate across.

5.4 RP: minimizes vs distributes

RP *minimizes* B_{MM} (it is gradient flow). In the multi-sector case, the minimum is achieved by *distributing* eliminable mismatch across sectors. “Distributes” is a consequence of “minimizes” in the multi-sector case, not an alternative. The irreducible floor $F^{(0)}$ is state-independent and cannot be distributed further. The single-sector case has nothing to distribute across, but minimization still happens: $F^{(1)} \rightarrow 0$ is the local minimum, with $F^{(0)}$ surviving.

6 Dark-Energy Fingerprint

6.1 The structural prediction

The tilted-Pauli split gives a clean structural prediction:

If the universe exhibits ongoing reconciliation dynamics (observed as cosmic expansion, structure formation, and $\rho_{DE} > 0$), then the universe’s local constraint set cannot be pure Pauli. The constraint tilt θ must be non-zero.

The logical chain:

1. Observed dynamics $\Rightarrow \hat{F}_{MM} \neq 0$ (RP flow requires non-zero mismatch to drive).
2. $\hat{F}_{MM} = 0$ identically for pure Pauli constraints ($\theta = 0$).
3. Therefore the constraints are non-Pauli: $\theta \neq 0$.

6.2 What is established

- The algebraic fact: $\hat{F}_{MM}^{(0)}(\theta) > 0$ for $\theta \in (0, \pi/2)$, and $\hat{F}_{MM}^{(0)}(0) = 0$. This is exact and verified.
- The structural form of the prediction: non-Pauli constraints are required for dynamics. This is sound.

- The fingerprint is a constraint on any future first-principles derivation of the constraint algebra: such a derivation must produce $\theta \neq 0$. A Pauli derivation would predict $\rho_{DE} = 0$, contradicting observation.

6.3 What is conditional

The empirical form of the prediction — that the *observed* ρ_{DE} mathematically forces $\theta \neq 0$ in the underlying algebra — is strictly **CONDITIONAL**. It requires the dimensional bridge mapping the algebraic expectation value $\hat{F}_{MM}^{(0)}$ to the macroscopic cosmological density ρ_{DE} . This bridge remains blocked by:

- **T-2** (spectral gap / infinite-dimensional convergence): the RP flow must be well-defined on the full (possibly infinite-dimensional) algebra.
- **T-3** (continuum limit): the discrete link structure must admit a continuum limit in which $\hat{F}_{MM}^{(0)}$ becomes an energy density.
- The Friedmann derivation (GAP-6.2.2): the cosmological equations must be derived from RP, not imported from standard GR.

Until T-2, T-3, and the Friedmann derivation are complete, the dark-energy fingerprint remains a **structural prediction**, not an established empirical fact.

6.4 What is not claimed

- The specific value of θ is *not* derived. It requires the Friedmann derivation and numerical simulation.
- The numerical amplitude of ρ_{DE} is *not* derived. The relation $\rho_{DE} \sim (H/\Gamma)^2 \cdot \hat{F}_{MM}^{(0)}(\theta)$ is structurally motivated but not dimensionally established.
- The fluctuation spectrum $(\delta T/T, n_s, r)$ is *not* derived. The earlier $n_s \approx 0.667$ prediction is retracted (Section 8).
- The identification of $\hat{F}_{MM}^{(0),\text{mode}}$ as the dark-energy source is *not* correct. Dark energy lives in the cross-sector channel ($\hat{F}_{MM}^{(0),\text{cross}}$), not the mode channel. The tilted-Pauli calculation is mode-channel (intra-sector) and sources particle masses via $m^2 \equiv B_0$, not dark energy. See Section 5 for the full single-sector vs multi-sector distinction.
- The empirical claim “observed ρ_{DE} forces non-Pauli constraints” is *not* established. It requires the multi-sector cross-channel computation ($\hat{F}_{MM}^{(0),\text{cross}}$), which is open.

6.5 Status

Status: Structural prediction, conditional on T-2/T-3 and the Friedmann derivation. The algebraic chain is sound. The empirical claim requires the dimensional connection, which is open.

7 Honest Status of the Hierarchical Proof

7.1 The hierarchical principle

The layer-hierarchical RP principle states that RP flow decomposes into layer-sequential constrained minimization:

1. **Layer Q'**: $\hat{F}_{MM}^{(1)}[\rho] = 0$ constrains the state.
2. **Layer C**: $B_{\text{geom}} = \text{Var}(g^{FS})$ is minimized within that constraint.
3. **Layer Q**: $B_{\text{causal}} = \sum \Phi_{\alpha\beta}$ is minimized within both.

If this holds, homogeneity follows: $\hat{F}_{MM}^{(1)} = \vec{W} \cdot \vec{r} = 0$ constrains $\vec{r}$ to a plane, and $B_{\text{geom}} \propto |\vec{r}|^4$ is uniquely minimized at $\vec{r} = 0$.

Scoping note. The above is a *single-sector local* statement. Cosmological homogeneity is a *multi-sector* statement: all sectors reach the same state $\vec{r}_k = \vec{r}_*$ for all k . This requires the sector field $\{\vec{r}_k\}$ and the cross-sector coupling, which the single-sector proof does not address. See Section 5.

7.2 The gap

The acyclic dependency graph (v3.0 §8.7, Thm 8.4) establishes that the emergence chain $L \rightarrow Q \rightarrow C \rightarrow Q'$ is *definitionally* acyclic. This is a fact about how definitions are ordered. It does **not** establish that RP flow *dynamically* respects this ordering. The bridge from “definitional acyclicity” to “dynamical layer-sequential minimization” has not been built. The critic correctly identified this: “a fact about how definitions are ordered on the page, and nothing connects definitional ordering to which point a gradient flow settles at.”

The most promising route is the Hessian identity (v3.0 Lemma 4.5): $g^{FS} = \frac{1}{2} \text{Hess}(\Phi)$. This gives a natural “constraint first, then measure” ordering — the metric is the Hessian of the log potential, so B_{geom} is only well-posed after $\hat{F}_{MM}^{(1)} = 0$ is satisfied. But this argument has not been formalized as a theorem.

7.3 Consequences for the primitive count

If the hierarchical principle is **proven**: the inflation-replacement requirement becomes derived, and the primitive count drops from 5 to 4. If **not proven**: the inflation-replacement requirement remains a primitive, and the count stays at 5.

Status: Primitive count is 5, conditional on 4 if the hierarchical principle is proven.

7.4 Future direction: hydrostatic-equilibrium framing

An intuitive framing: in a stratified fluid, pressure equilibrates sequentially from the top layer down, because each layer’s equilibrium depends on boundary conditions set by the layer above. If RP flow on the burden “fluid” behaves analogously, the hierarchical principle would follow from burden conservation on a stratified state manifold. This is a *research direction*, not a derivation. The formal theorem has not been written. Noted here as a guide for future work.

8 Retraction Register and Updated Scorecard

8.1 Retracted claims

Retracted claim	Reason
Variance-form B_{MM}^{var}	Not derived from primitives; contradicts homogeneity.
Mexican-hat / SSB framing	Contradiction dressed as feature.
$n_s \approx 0.667$ prediction	Arithmetically wrong ($n_s = 2/3$, off by 0.3 from 0.965).
Numerical $ \tilde{r} _*$ values	Three values, none verified.
$\hat{F}_{MM}^{(0),\text{mode}}$ as DE source	Category error; DE is cross-sector.
$E = mc^2$ from “resistance”	Lorentz-blocked; scaling guess, not derivation.
“Three seeds of mismatch”	Jacobi is identity; Kaplansky is eligibility.
Bounded degree from B_{causal}	Plausible, not derived.
Mass = “resistance”	Replaced by displacement vs. resistance distinction.
Global cosmological overextensions	Single-sector tilted-Pauli interpreted as global DE / homogeneity / fluctuations; requires multi-sector (Sec. 5).
“Homogeneity uncond. derived”	Conditional on hierarchical principle (OPEN); also needs multi-sector extension.
“Primitive count is 4”	Conditional; if unproven, count is 5.

8.2 Consolidated scorecard

Claim	Status
<i>Established</i>	
$\hat{F}_{MM} = \sum \omega(J^2) \cdot \rho(\Delta^\dagger \Delta)$	Established
Pauli-zero as fully reconciled limit	Established (robust under multi-sector extension)
Tilted-Pauli $\hat{F}_{MM}^{(0)}/\hat{F}_{MM}^{(1)}$ split	Derived (intra-sector; critic-verified)
Bures metric selection (Petz + Cramér–Rao)	Established
Two-Link derivation (Leibniz identity)	Established
<i>Structural prediction (T-2/T-3-blocked empirically)</i>	
Dark-energy fingerprint ($\theta \neq 0$)	Structural; empirical T-2/T-3-blocked
<i>Conditional (on unproven hierarchical principle)</i>	
Hierarchical homogeneity proof	OPEN
Inflation-replacement primitive elimination	Conditional
Primitive count 4 (vs. 5)	Conditional
<i>Open</i>	
T-2 (spectral gap + bounded degree)	OPEN (existential)
T-3 (continuum limit)	OPEN
Friedmann derivation (GAP-6.2.2)	OPEN
$\hat{F}_{MM}^{(0)} \rightarrow \rho_{DE}$ dimensional connection	OPEN
Fluctuation spectrum ($\delta T/T, n_s, r$)	OPEN
Bounded degree from $B_{\text{causal}}^{\text{mode}}$	OPEN (plausible)
Multi-sector $\hat{F}_{MM}^{(0),\text{cross}}$	OPEN
Mass from burden displacement	OPEN (research program)
$E = mc^2$ from RP	OPEN (Lorentz-blocked)
<i>Retracted</i>	
Variance-form, Mexican-hat, $n_s \approx 0.667$, etc.	Retracted (see above)

8.3 Net assessment

The framework’s foundations are sound: $\omega(J^2)$ construction, Pauli-zero (robust under multi-sector extension), tilted-Pauli split (intra-sector), and structural dark-energy fingerprint are preserved. The scoping correction (§5) clarifies local vs. global.

Major Revision (unchanged). Highest-leverage open targets: (1) layer-hierarchical RP principle (closes homogeneity, primitive count $\rightarrow 4$); (2) T-2 / bounded degree (closes 12 of 26 gaps). Both well-posed.